

WEAR OR EROSION?

TUBE-TO-TUBE WEAR VS. EROSION

Two kinds of wall loss that can look alike. Location and the neighboring tube usually settle it.

KEY IDEA

Ask two things: where is it, and does the tube next door have the same thing at the same spot?

	Tube-to-tube wear	Erosion
WHERE	Free span between supports, often mid-span or in the U-bend	Usually the inlet end, just past the tube sheet. Sometimes near the shell inlet or impingement area
ID OR OD	OD: tubes rub on each other's outside	Inlet erosion is usually ID. Impingement erosion is OD
SHAPE	Fairly localized: a defined spot or short length	Gradual and long: a slow drift on the absolute
NEIGHBOR TUBE	Matching wear at the same spot	No matching partner. Many tubes in the same region
LOW FREQUENCY	Often a proximity signal from close tubes	Nothing special

1 LOCATION FIRST

- Inlet end near the tube sheet: think erosion.
- Free span or U-bend, away from supports: think tube-to-tube wear.

2 CHECK THE NEIGHBOR

STRONGEST TEST

- Tube-to-tube wear takes two tubes, so both should show wall loss at the same spot.
- If the neighbor is clean there, tube-to-tube wear is unlikely.

3 ID OR OD, SHAPE, PROXIMITY

- ID signals are strongest on CH1. Tube-to-tube wear is always OD.
- Erosion starts gradually and fades out. Tube-to-tube wear is a defined spot.
- A proximity signal on the low-frequency absolute supports tube-to-tube wear.

THE TRICKY CASE

- › OD erosion near the shell inlet can sit in the free span and look like tube-to-tube wear
- › Partner tube plus proximity signal: tube-to-tube wear
- › No partner, broad area across tubes facing the inlet: erosion

BUNDLE PATTERN

- › Erosion: many tubes in the same region
- › Tube-to-tube: pairs of adjacent tubes

QUICK TIPS

Always look up the adjacent tube at the same elevation.

Your site may code tube-to-tube wear separately. Check your list.

Can't tell which it is? Flag it.